Computer Science & IT



C Programming

Data type & Operator

Lecture No. 05



By- Abhishek Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

++ , -- operator (Imp)

Relational operator

Slide

Topics to be Covered



Topic Logical operator

Topic && AND

Topic || logical OR

Topic ! logical Not

Topic Bitwise operator

Slide



topic: Logical Operator

a	b	a && b
0	0	0
0	1	0
1	0	0
1	1	1

if one operator is 0 then Answer is zero

a	b	a || b
0	0	0
0	1	1
1	0	1
1	1	1

if one operator is 1 then Ans is 1

! ← unary operator

a	!a
0	0 → 1
1	1 → 0

Slide



topic: Logical Operator (AND) &&

```
#include <stdio.h>
int main(void) {
    int a = 20;
    int b = 30;
    printf("%d", a > 10 && b > 10);
}
```

20 > 10 && 30 > 10

1 && 1 = 1

Slide



Topic: Logical Operator (OR) ||

```
#include <stdio.h>
int main(void) {
   int a = 20;
   int b = 5;
   printf("%d", a > 10 || b > 10);

}
```

20>10 || 5>10

1 || 0 : 1

Slide



Topic: Logical Operator (NOT) !

```c
#include <stdio.h>
int main(void) {
   int a = 20;
   int b = 5;
   printf("%d", !(a < 10 ));
}
```

C-Language

Nonzero ⟶ true - 1

zero — false — 0

!(20<10)

!(0) = 1

printf (!a < 10)

!1 < 10

0 < 10

true

Slide



Topic: Precedence

3	* / %	Multiplication, division, and modulus	left to right
4	+ -	Addition and subtraction	left to right
6	< <= > >=	Relational less than and less than or equal to Relational greater than and greater than or equal to	left to right
7	== !=	Relational equal to and not equal to	left to right
11	&&	Logical AND	left to right
12	||	Logical OR	left to right



Topic: Question

```
#include<stdio.h>
int main(){
      int x = 1, a;

      a = x || ++x
      printf("%d", x);


      return 0;
}
```

The value printed is [1]

Logical expression

precedence rule dont apply

Short circuit code

1. left to right evaluation

1 || ++x

Rest of expression will Not evaluated

Slide



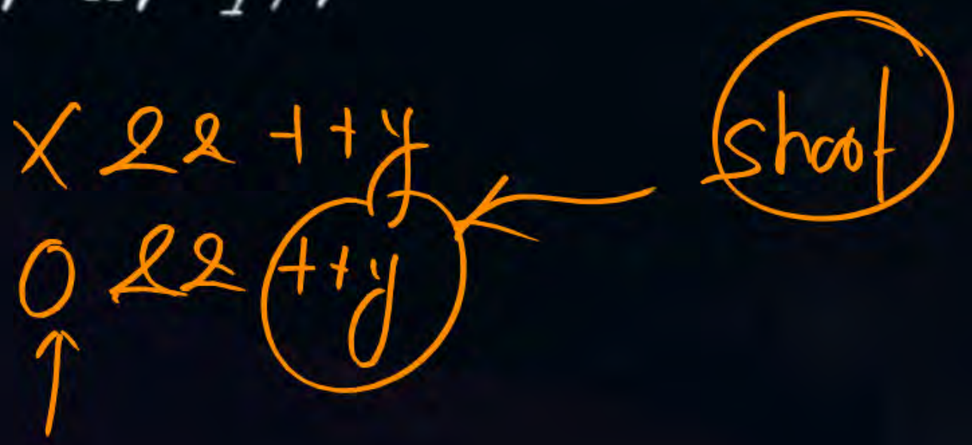
Topic: Question

```
#include<stdio.h>
int main(){
    int x = 0, y = 0, a;

    a = x && ++y;
    printf("%d %d", x, y);

    return 0;
}
```

A. 1 1
B. 1 0
C. 0 1
D. 0 0



Slide



Topic: Question

```
#include<stdio.h>
int main(){
    int x = 0, y = 0, a;

    a = x && ++y;
    printf("%d %d", x, y);


    return 0;
}
```

A. 1 1
B. 1 0
C. 0 1
D. 0 0

Slide

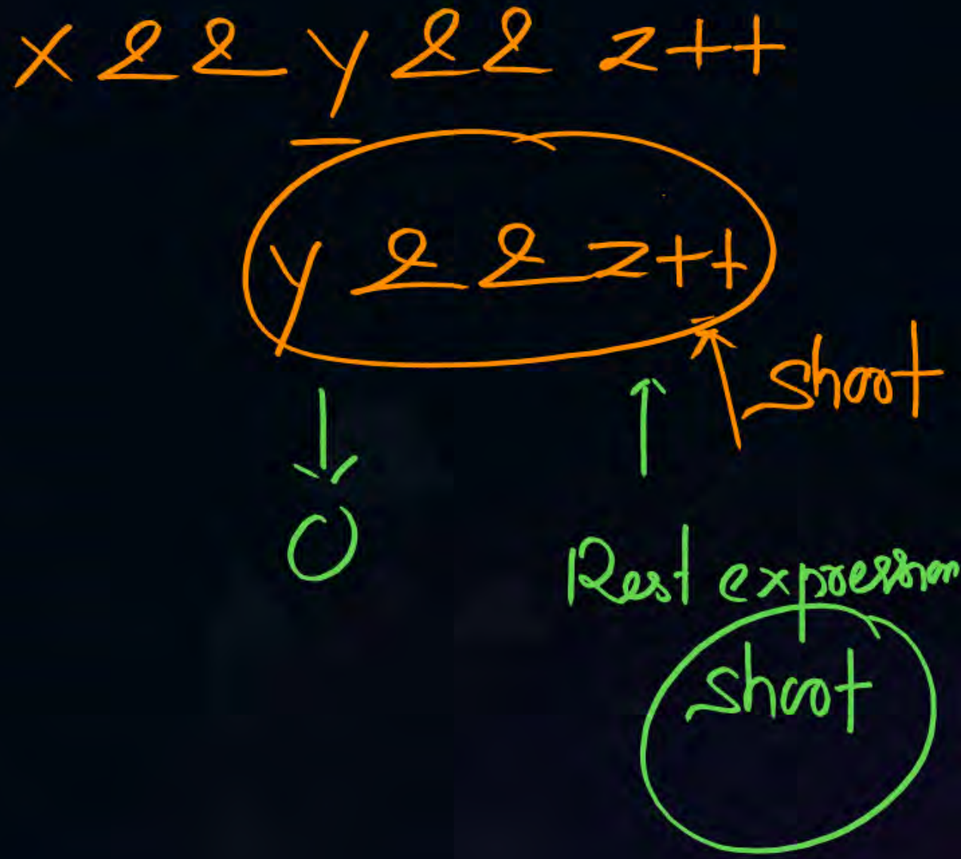


Topic: Question

```
#include <stdio.h>
    void main ()  {
        int x = 1, y = 0, z = 5;
        int a = x && y && z++;
        printf("%d", z);
    }
```

The value printed is ?

A. 6
B. 5
C. 0
D. 1



Slide

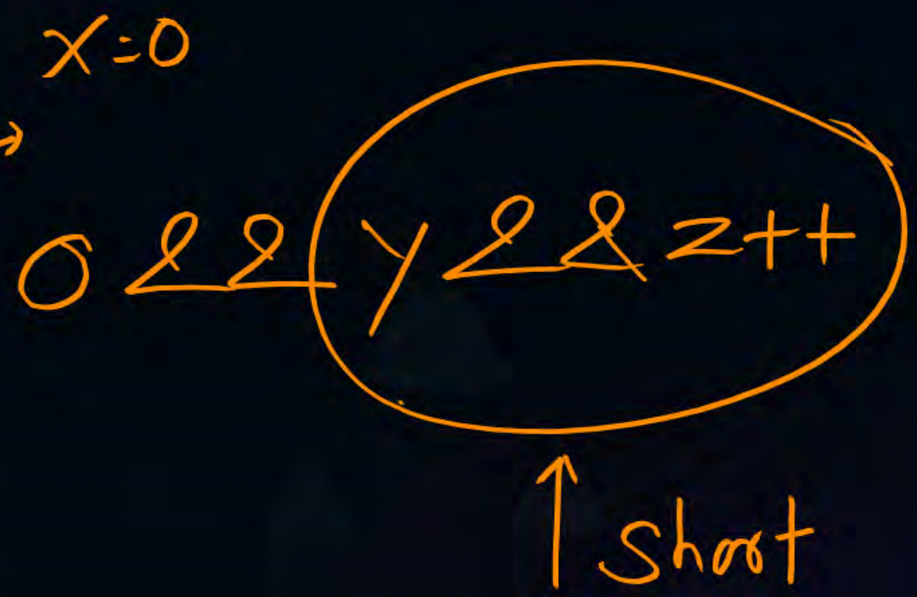


Topic: Question

```
#include <stdio.h>
    void main ()  {
        int x = 1, y = 0, z = 5;
        int a = x && y && z++;
        printf("%d", z);
    }
```

The value printed is ?

A. 6
B. 5
C. 0
D. 1



Slide



Topic: Short Circuit Code

```
#include <stdio.h>
void main ()  {
      int x = 1, y = 0, z = 0;
      int a = x++ || y++ && z++;
      printf("%d, %d",y, z);
}
```

A. 1,1
B. 1,0
C. 0,1
D. 0,0

Slide

Short

x++ || y++ && z++

1 || 0 && 0

1 || 0 = 1

1

1 || 0 = 1

regardless of value of expression (0,1)

the final Answer 1



Topic: Short Circuit Code

```
#include <stdio.h>
void main ()  {
      int x = 1, y = 0, z = 0;
      int a = x++ || y++ && z++;
      printf("%d, %d",y, z);
}
```

A. 1,1
B. 1,0
C. 0,1
D. 0,0

Slide



Topic: Short Circuit Code

```
#include <stdio.h>
int main ()  {
      int x = 0, y = 0, z = 0;
      int a = x++ || y++ && z++;
      printf("%d, %d",y, z);
      return 0;
}
```

A. 1,1
B. 1,0
C. 0,1
D. 0,0

Slide

Bit-wise Operator



		R
2	10	
2	5	0
2	2	1
2	1	0
	0	1

1010

Binay representation

1111 = 15 decimal

1010 = 10 decimal

Slide

Number System



		Remainder
2 | 24		
2 | 12		0
2 | 6		0
2 | 3		0
2 | 1		1
	0	1

11000

		Remainder
2 | 37		
2 | 18		1
2 | 9		0
2 | 4		1
2 | 2		0
2 | 1		0
	0	1

100101

$5\ 4\ 3\ 2\ 1\ 0$

Slide

47

32+15

1 0 1 1 1 1
5 4 3 2 1 0

32+15

113

64+32+16+1·+2⁰

1110001



Bit-wise Operator

Operators	Meaning of operators
&	Bitwise AND
|	Bitwise OR
^	Bitwise exclusive OR
~	Bitwise complement
<<	Shift left
>>	Shift right

Slide



Bit-wise Operator

What is the output the program

```
#include <stdio.h>
    int main ()  {
      int x = 5, y=17, z
      z = x&y;
      printf("%d", z);
 }
```

(A) 1 (B) 21

(C) 2 (D) -6



Bit-wise Operator

What is the output the program

```
#include <stdio.h>
    int main ()  {
      int x = 5, y=17, z
      z = x&y;
      printf("%d", z);
 }
```

(A) 1

(B) 21

(C) 2

(D) -6

X 00101

Y 10001

00001

AND

2 | 17	
2 | 8	1
2 | 4	0
2 | 2	0
2 | 1	0
	0 1

Slide



Bit-wise Operator

What is the output the program

```
#include <stdio.h>
    int main ()  {
       int x = 5, y=17, z
       z = x|y;
       printf("%d", z);
 }
```

(A) 1 (B) 21

(C) 2 (D) -6

x = 00101

y = 10001

10101.

16+5 = 21

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Bit-wise Operator

What is the output the program

```
#include <stdio.h>
    int main ()  {
      int x = 5, y=17, z
      z = x|y;
      printf("%d", z);
 }
```

(A) 1 **(B) 21**

(C) 2 (D) -6

Slide



Bit-wise Operator

What is the output the program

```
#include <stdio.h>
    int main ()  {
       int x = 5, y=17, z
       z = x^y;
       printf("%d", z);
 }
```

(A) 1
(B) 21
(C) 20
(D) -6

a	b	a^b
0	0	0
0	1	1
1	0	1
1	1	0

XOR
exclusive
OR

X = 00101

Y = 10001

10100 = 20

Slide

1010

$1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 \times 0 \times 2^0$

$= 10$ unsigned



Bit-wise Operator

1 - ve 0 - +ve

What is the output the program

```
#include <stdio.h>
    int main ()  {
        int x = 5, z
        z = ~x;
        printf("%d", z);
    }
```

(A) 1
(B) 21
(C) 20
(D) -6

~ Bitwise Negation

0 101

~(0101)

3 2 1 0

= 1010

$-2^3 + 2^1 = -8+2 = -6$

2's complement Interpretation

take extra bit for +ve 0 as extra Sign

Slide

2's complement

+5

↓

0101

1's complement ⟶ 1010

addition 1 → 1

1011 (−5)

3210

$-2^3 + 1\times2^1 + 1\times2^0 = -8+3 = -5$



Bit-wise Operator

What is the output the program

```
#include <stdio.h>
    int main ()  {
        int x = 5, z
        z = ~x;
        printf("%d", z);
    }
```

(A) 1
(B) 21
(C) 20
(D) -6

$$\sim x = -(x+1)$$

$X = 250 : -(250+1)$

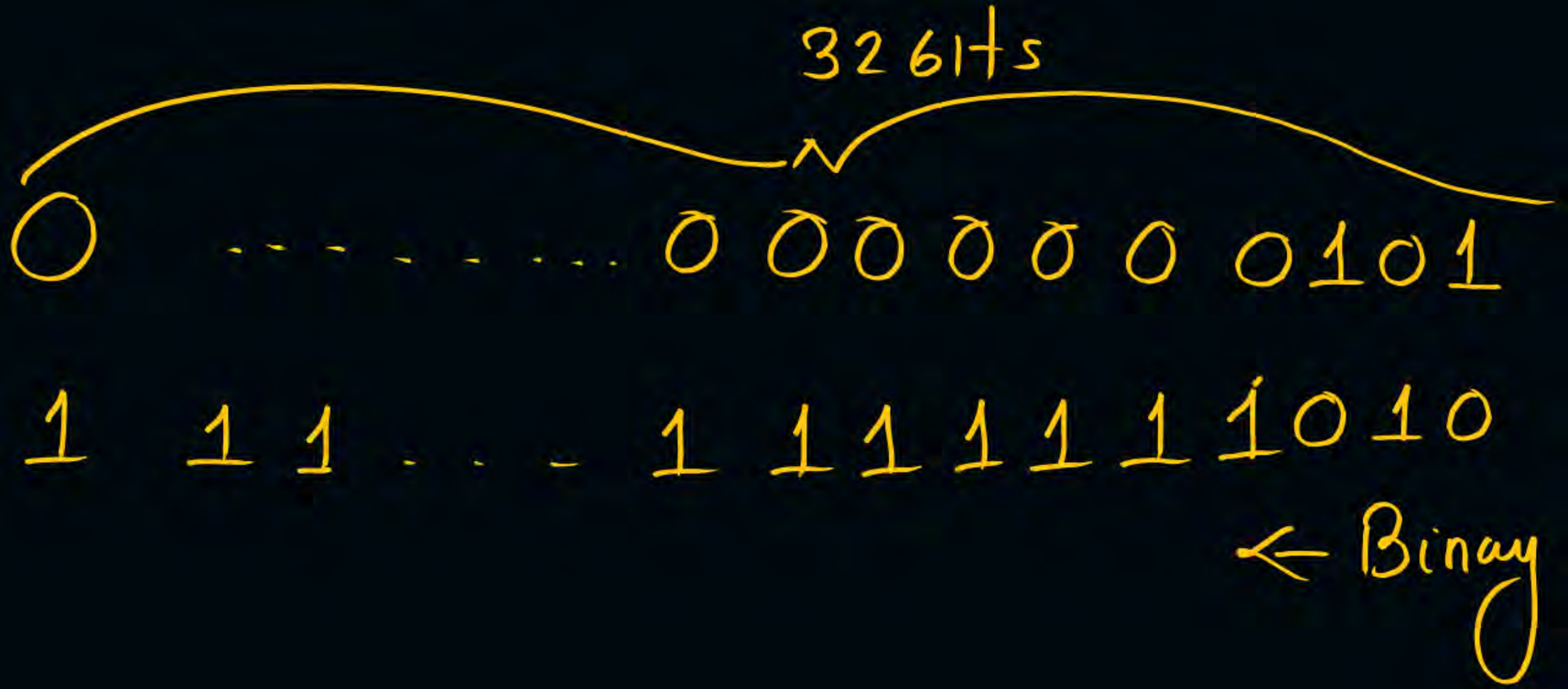
$: -251$

$X = -250 =$

$\sim X = -(-250+1)$

$-(-249) = 249$

Slide


32 bits
0 00000005
1 11 ... 1111111A
← Binary, %u



2 mins Summary

Topic	Logical operator
Topic	AND, OR, NOT
Topic	Short circuit code
Topic	bit AND, bitwise OR. XOR. Negation
Topic	

Slide

THANK - YOU

